

Topics

1. CIIT/FA20-BCS-019/SWL
2. CIIT/FA21-BCS-003/SWL
3. CIIT/FA21-BCS-004/SWL
4. CIIT/FA21-BCS-005/SWL
5. CIIT/FA21-BCS-008/SWL
6. CIIT/FA21-BCS-009/SWL
7. CIIT/FA21-BCS-014/SWL
8. CIIT/FA21-BCS-015/SWL
9. CIIT/FA21-BCS-017/SWL
10. CIIT/FA21-BCS-018/SWL
11. CIIT/FA21-BCS-019/SWL
12. CIIT/FA21-BCS-020/SWL
13. CIIT/FA21-BCS-021/SWL
14. CIIT/FA21-BCS-024/SWL
15. CIIT/FA21-BCS-026/SWL
16. CIIT/FA21-BCS-027/SWL
17. CIIT/FA21-BCS-029/SWL
18. CIIT/FA21-BCS-030/SWL
19. CIIT/FA21-BCS-031/SWL
20. CIIT/FA21-BCS-034/SWL
21. CIIT/FA21-BCS-035/SWL

22. CIIT/FA21-BCS-037/SWL
23. CIIT/FA21-BCS-038/SWL
24. CIIT/FA21-BCS-039/SWL
25. CIIT/FA21-BCS-041/SWL
26. CIIT/FA21-BCS-042/SWL
27. CIIT/FA21-BCS-044/SWL
28. CIIT/FA21-BCS-046/SWL
29. CIIT/FA21-BCS-049/SWL
30. CIIT/FA21-BCS-050/SWL
31. CIIT/FA21-BCS-152/SWL
32. CIIT/FA21-BCS-154/SWL
33. CIIT/FA21-BCS-155/SWL
34. CIIT/FA21-BCS-159/SWL
35. CIIT/FA21-BCS-160/SWL
36. CIIT/FA21-BCS-161/SWL
37. CIIT/FA21-BCS-164/SWL
38. CIIT/FA21-BCS-165/SWL
39. CIIT/FA21-BCS-166/SWL
40. CIIT/FA21-BCS-167/SWL
41. CIIT/FA21-BCS-168/SWL
42. CIIT/FA21-BCS-178/SWL
43. CIIT/FA21-BCS-223/SWL

1. Introduction to Genome Data Science

- 1.1. Definition and scope of bioinformatics
- 1.2. Importance of data in modern biology
- 1.3. Overview of genome data science
- 1.4. Role of Python in bioinformatics

2. Molecular Biology for Computer Scientists

- 2.1. Structure and function of DNA, RNA, proteins
- 2.2. Genes and chromosomes
- 2.3. Cell types and gene regulation

3. Central Dogma: DNA Replication, Transcription, Translation

- 3.1. DNA replication mechanisms
- 3.2. RNA transcription and processing
- 3.3. Protein translation and folding

4. Gene Expression Mechanisms

- 4.1. Gene regulation
- 4.2. Epigenetic modifications
- 4.3. Promoters, enhancers, transcription factors

5. Omics Overview: Genomics, Transcriptomics, Proteomics

- 5.1. Genomics: whole genome analysis
- 5.2. Transcriptomics: mRNA expression
- 5.3. Proteomics: protein profiling and interactions

6. Genome Sequencing Technologies

- 6.1. Sanger sequencing
- 6.2. Next-generation sequencing (Illumina, Ion Torrent)
- 6.3. Third-generation sequencing (PacBio, Oxford Nanopore)

7. Biological Databases

7.1.NCBI, Ensembl, UniProt

7.2.Database querying and access methods

7.3.Importance of data repositories

8. Genomic File Formats

8.1.FASTA, FASTQ, GFF/GTF, SAM/BAM, VCF

8.2.Structure and purpose of each format

8.3.Tools for parsing and analyzing file formats

9. Python Programming for Bioinformatics

9.1.Variables, loops, conditionals

9.2.Functions and modules

9.3.Python scripting for bio-tasks

10.Data Structures in Python

10.1. Lists, tuples, dictionaries

10.2. Nested data structures

10.3. Applications in biological datasets

11.Data Acquisition Techniques

11.1. Web scraping for biological data

11.2. APIs: NCBI Entrez, EBI

11.3. Handling large datasets

12.Data Preprocessing and Cleaning

- 12.1. Removing missing values
- 12.2. Normalization and scaling
- 12.3. Filtering and transformation

13.Sequence File Parsing

- 13.1. Reading FASTA/FASTQ using Biopython
- 13.2. Extracting headers, sequences, and quality scores
- 13.3. Writing parsed data to structured files

14.Using Bioinformatics Libraries

- 14.1. Biopython, scikit-bio
- 14.2. pandas, NumPy
- 14.3. Bioinformatics-specific functions

15.Data Visualization Techniques

- 15.1. Line graphs, bar charts, scatter plots
- 15.2. Biological heatmaps, sequence logos
- 15.3. Libraries: matplotlib, seaborn, plotly

16.Reporting and Dashboards

- 16.1. Jupyter Notebooks
- 16.2. Streamlit for interactive bio-apps
- 16.3. Exporting results to HTML, PDF

17.Sequence Alignment Algorithms

- 17.1. Global (Needleman-Wunsch)
- 17.2. Local (Smith-Waterman)
- 17.3. Scoring matrices (PAM, BLOSUM)

18.Multiple Sequence Alignment

- 18.1. Algorithms: ClustalW, MUSCLE, MAFFT
- 18.2. Consensus sequence generation
- 18.3. Applications in phylogeny and motif discovery

19.Phylogenetic Tree Construction

- 19.1. Types: rooted vs. unrooted
- 19.2. Methods: UPGMA, NJ, ML, Bayesian
- 19.3. Interpreting evolutionary relationships

20.Genome Assembly Principles

- 20.1. Overlap-layout-consensus (OLC)
- 20.2. de Bruijn graphs
- 20.3. Contigs and scaffolding

21.Variant Calling and Analysis

- 21.1. SNPs, indels, structural variants
- 21.2. Tools: BWA, GATK, SAMtools
- 21.3. Annotation of variants

22.Functional Annotation and GO

- 22.1. Gene Ontology (GO) categories
- 22.2. Annotation pipelines
- 22.3. Enrichment analysis

23.Transcriptomics and RNA-Seq

- 23.1. RNA extraction and sequencing
- 23.2. Expression quantification (TPM, FPKM)
- 23.3. Differential expression analysis**

24.Epigenomics and Regulatory Genomics

- 24.1. DNA methylation and histone modification
- 24.2. ChIP-Seq and ATAC-Seq
- 24.3. Role in gene regulation

25.Statistics in Genomics

- 25.1. Hypothesis testing
- 25.2. P-values, FDR
- 25.3. Correlation and regression

26.Statistical Tests for Bioinformatics

- 26.1. t-test, ANOVA, chi-square
- 26.2. Non-parametric methods
- 26.3. Use cases in expression analysis

27. Machine Learning Basics

- 27.1. Supervised vs. unsupervised learning
- 27.2. Model training and testing
- 27.3. Use of scikit-learn

28. Feature Extraction in Biological Data

- 28.1. One-hot encoding for sequences
- 28.2. K-mer counting
- 28.3. Dimensionality reduction (PCA, t-SNE)

29. Classification and Clustering

- 29.1. k-NN, SVM, Random Forests
- 29.2. K-means, hierarchical clustering
- 29.3. Evaluation metrics (accuracy, AUC)

30. Advanced ML & Deep Learning

- 30.1. Neural networks
- 30.2. CNNs for image/genomic data
- 30.3. Deep learning for sequence analysis

31. GWAS and Association Studies

- 31.1. SNP-trait associations
- 31.2. Population structure correction
- 31.3. Visualization: Manhattan plots

32. Protein Structure Prediction

- 32.1. Primary to quaternary structure
- 32.2. Homology modeling, threading
- 32.3. Tools: SWISS-MODEL, AlphaFold

33.Molecular Docking

- 33.1. Ligand-receptor interaction
- 33.2. Docking tools: AutoDock, PyMOL
- 33.3. Scoring functions

34.Systems Biology

- 34.1. Biological networks and interactions
- 34.2. Pathway modeling
- 34.3. KEGG, Reactome databases

35.Pathway Analysis

- 35.1. Identifying enriched pathways
- 35.2. Functional implications of gene sets
- 35.3. Visualization tools (Cytoscape)

36.Multi-Omics Integration

- 36.1. Combining genomics, transcriptomics, proteomics
- 36.2. Correlation and co-expression analysis
- 36.3. Network-based integration

37.Big Data in Genomics

- 37.1. Storage and management of large datasets
- 37.2. Distributed computing
- 37.3. Introduction to Hadoop/Spark

38.Cloud Computing for Bioinformatics

- 38.1. AWS, GCP for genomics pipelines
- 38.2. Workflow management tools: Galaxy, Nextflow
- 38.3. Scalability and reproducibility

39.Literature Review Techniques

- 39.1. PubMed and Google Scholar search
- 39.2. Identifying research gaps
- 39.3. Evaluating scientific rigor

40.Project Design in Bioinformatics

- 40.1. Defining problem statement
- 40.2. Data collection and analysis
- 40.3. Evaluation and interpretation

41.Ethical, Legal & Social Issues (ELSI)

- 41.1. Privacy of genomic data
- 41.2. Genetic discrimination
- 41.3. Consent and data sharing

42.Reproducible Research Practices

- 42.1. Version control (Git/GitHub)
- 42.2. Documenting code and workflows
- 42.3. Sharing datasets and results

43.Bioinformatics Careers and Research

- 43.1. Career paths: academia, industry, clinical
- 43.2. Current research areas
- 43.3. Building a portfolio